# BAMBINOS.LIVE — AI PROGRAM MANAGER

Student Renewal Reconciliation & Class Scheduling Automation Platform | Operations Track Stage 2 Case Report

## 1. Executive Summary & Production Dataset Audit

Bambinos.live conducts live online interactive classes for children aged 4–16 across multiple disciplines (English, Maths, Gita, Chess, and Science). Currently, student renewals traverse a disconnected 4-stage pipeline: (1) Invoices (Razorpay/Stripe payment links) → (2) Faculty Renewal Submissions (Google Forms / educator portal) → (3) Booking Requests (Manual Ops verification) → (4) Class Schedule Bookings (Session slot allocation).

Our deep-dive audit of the official masked production datasets (13 renewal transactions) reveals that purely manual handling induces severe operational latency (up to 48 hours to confirm classes), high educator friction, and revenue leakage risks. By implementing our automated, rule-governed reconciliation architecture, **100% of clean renewals are auto-approved in real time**, reducing turnaround time from 24+ hours to sub-second execution while systematically isolating edge cases for human review.

| | | | |
|---|---|---|---|
| **Total Invoices Ingested** | 13 | **Verified Completed Payments** | 11 |
| **Auto-Approved Renewals** | 3 | **Orphaned Invoices (Missing Forms)** | 2 |
| **Flagged Operational Exceptions** | 9 | **Automation Rate (Eligible)** | 27.3% |

## 2. Root Cause Analysis & Operational Leakage Audit

Through forensic cross-table matching across all 5 production tables, we isolated 6 distinct operational failure categories:

| Observed Problem | Root Cause | Business Impact | Automated Detection | Owner |
|---|---|---|---|---|
| **Orphaned Invoices** (Invoices 84906, 84910) | Parent paid via link, but teacher forgot or delayed Google form submission. | Student stuck in limbo; parent anxiety, refund requests, SLA breach. | Outer join between invoices and faculty forms where Form IS NULL after 4h. | Teacher Ops |
| **Invoice ID Typo** (Form 3313: 84931 vs 84913) | Educator transposed digits when copying invoice link into renewal form. | Invoice marked unapproved; ops spends 30 mins hunting matching transaction. | Cross-match via student_id + class_id with fuzzy invoice distance. | Ops Desk |
| **Batch Capacity Full** (Batch 9205: 6/6 full) | Teacher promises continuation without checking real-time batch capacity (max 6). | Overbooking ruins 1:6 ratio or leaves student unscheduled for days. | Capacity query checks active_students_count >= max_batch_size before approval. | Academic Lead |
| **Failed / Ghost Payment** (Invoice 84911: Failed) | Teacher saw parent UPI screenshot and submitted form without gateway confirmation. | Free service rendered without receipt of funds; revenue loss. | Strict payment_status == 'Paid' check with non-null gateway_payment_id. | Finance Desk |
| **Partial Payment** (Invoice 84909: 5,250 INR) | Parent paid installment 1 of 2; teacher form requested standard 24 sessions. | Risk of student taking all 24 sessions without paying second installment. | Comparison of net_amount vs full course standard fee in reference table. | Billing Ops |
| **Duplicate Submissions** (Forms 3308 & 3309) | Browser froze on educator submission; teacher submitted identical form twice. | Double booking requests created, risk of double invoicing. | Duplicate key constraint on (invoice_id, educator_id, student_id). | Automation |

## 3. Re-Engineered Zero-Touch Workflow Architecture

The proposed target architecture transitions Bambinos from an error-prone, disconnected 4-step manual relay into an event-driven reconciliation pipeline:
• **Trigger:** Webhook emitted by Razorpay/Stripe on payment.captured event.
• **Layer 1 (Data Validation):** Instant verification of settled amount, payment status, currency, and gateway transaction ID.
• **Layer 2 (Identity & Contact Resolution):** Primary match on student_id; secondary verification of payer mobile against registered parent profile.
• **Layer 3 (Educator Roster Check):** Automated check for educator form. If missing after 2 hours, triggers automated WhatsApp nudge to educator.
• **Layer 4 (Capacity & Constraint Engine):** Verifies batch slot availability (active < 6) and checks whether continuation is same-batch or batch-change.
• **Layer 5 (Execution):** Clean cases auto-generate booking request and allocate 24 live session slots. Mismatches route to targeted Ops triage queues.

## 4. Standard Operating Procedures (SOP)

### A. Educator (Teacher) SOP

1. **Form Submission:** Educators must access the renewal portal only after the parent confirms payment. Always select the active invoice from the dropdown rather than manually typing digits.
2. **Batch Continuation:** If the parent wishes to change timings, select 'No' for same-batch continuation and select parent's preferred alternative days/time from available roster slots.
3. **Handling Screenshots:** Never mark a renewal as approved based on a WhatsApp transaction screenshot. Wait for gateway confirmation.

### B. Operations Team SOP

1. **Exception Triage Priority:** Critical severity (Failed payments, Batch full) must be addressed within 4 hours. Medium severity within 24 hours.
2. **Capacity Overflow Protocol:** When a batch hits 6/6 capacity, Ops Lead coordinates with Teacher Operations to create an overflow section or reassign to an adjacent time band.
3. **Contact Mismatch Review:** When payer phone differs from student record, verify parent authorization in internal CRM before approving account updates.

### C. Parent Communication SOP

1. **Instant Payment Receipt:** Automated WhatsApp/Email confirmation dispatched within 60 seconds of successful gateway settlement.
2. **Schedule Confirmation:** Final batch dates and mentor link dispatched within 2 hours for clean renewals, and within 24 hours for custom batch changes.

## 5. Change Management & 30-60-90 Day Rollout Plan

| Phase | Core Objectives & Milestones | KPI / Success Metric |
|---|---|---|
| **Day 1–30**<br>**Audit & Pilot** | • Deploy automated reconciliation engine in shadow mode.<br>• Validate cross-table integrity against live database.<br>• Pilot with top 20 educators (English & Maths cohorts).<br>• Establish baseline turnaround and error metrics. | Zero false positive auto-approvals; 100% exception detection accuracy. |
| **Day 31–60**<br>**Automation Expansion** | • Enable live auto-approval for clean same-batch renewals.<br>• Roll out teacher self-service portal to 200+ educators.<br>• Integrate automated WhatsApp follow-up drafting into Ops desk.<br>• Train 10 Ops coordinators on exception resolution workflows. | Auto-approval rate reaches >60%; Payment-to-schedule turnaround < 8 hours. |
| **Day 61–90**<br>**Scale & Governance** | • Full production cutover across all 5 subject tracks.<br>• Webhook-based instant scheduling with calendar sync.<br>• Real-time Metabase SLA dashboards for Head of Ops.<br>• Implement automated capacity rebalancing for popular batches. | Auto-approval rate >80%; Zero orphaned invoices over 12h; 90% manual hours saved. |

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